Med Term Quiz #2 — Immunity & Blood

Study guide for NRSG 231 Med Terminology Quiz #2 (due Oct 14, 2026). Focus: blood and immune/lymphatic terminology from the med-term textbook (cardiovascular/blood chapter; lymphatic/immune chapter) plus Pathophysiology terminology from Blood, Immune 1, and Immune 2 lectures. **Spelling counts.** Source tags: [handout] = on the course study-guide handout; [general] = standard medical terminology, not on that handout. Handout typos noted where relevant.

1. Word Parts

Blood & Immune Roots / Combining Forms

Part	Meaning	Example	Source
hem/o	blood	hemolysis	[general]
hemat/o	blood	hematocrit	[general]
erythr/o	red	erythrocyte	[handout] (listed as "Eryhr" — *typo; correct form is erythr/o*)
leuk/o	white	leukocyte	[handout]
thromb/o	clot; platelet	thrombocyte	[general]
cyt/o	cell	cytology	[general]
phleb/o	vein	phlebitis	[general]
ven/o	vein	venipuncture	[general]
lymph/o	lymph	lymphocyte	[general]
lymphaden/o	lymph node	lymphadenopathy	[general]
splen/o	spleen	splenomegaly	[general]
thym/o	thymus gland	thymectomy	[general]
tonsill/o	tonsils	tonsillitis	[general]
immun/o	immune; immunity	immunodeficiency	[handout]
path/o	disease	pathology	[general]
phag/o	to eat, devour	phagocyte	[general]
sider/o	iron	sideropenia	[general]
myel/o	bone marrow; also spinal cord	myelopoiesis	[handout] (handout lists "spinal cord"; in blood context = bone marrow)
plasm/o	plasma; formed substance	plasmapheresis	[general]
kary/o	nucleus	karyolysis	[general]
granul/o	granule	granulocyte	[general]
neutr/o	neutral (staining)	neutrophil	[general]
eosin/o	eosin (red dye); rose-colored	eosinophil	[general]
bas/o	base (basic-staining)	basophil	[general]

morph/o	shape; form	morphology	[general]
poikil/o	varied; irregular	poikilocytosis	[general]

Prefixes

Part	Meaning	Example	Source
aniso-	unequal; uneven	anisocytosis	[general]
an-	without; not	anemia	[handout]
pan-	all	pancytopenia	[general]
poly-	many; much	polycythemia	[general]
auto-	self	autoimmune	[general]
anti-	against	antibody	[general]
hyper-	excessive; above normal	hyperbilirubinemia	[handout]
hypo-	deficient; decreased	hypovolemia	[handout]
macro-	large; abnormal size	macrophage	[handout]
micro-	small	microcytic	[handout]

Suffixes

Part	Meaning	Example	Source
-cyte	cell	erythrocyte	[handout]
-emia	blood condition	anemia	[general]
-penia	deficiency; decrease	leukopenia	[general]
-poiesis	formation; production	hematopoiesis	[general]
-lysis	destruction; breakdown	hemolysis	[general]
-stasis	stopping; standing still	hemostasis	[general]
-osis	abnormal condition; disease	leukocytosis	[handout]
-philia	attraction to; affinity for	eosinophilia	[general]
-megaly	enlargement of	splenomegaly	[handout]
-pathy	disease	lymphadenopathy	[general]
-itis	inflammation of	phlebitis	[handout]
-oma	tumor of	lymphoma	[handout]
-rrhage	bursting forth; bleeding	hemorrhage	[handout]
-ectomy	surgical removal	splenectomy	[handout]
-tomy	cutting; incision	phlebotomy	[general]
-globin	protein (blood)	hemoglobin	[general]
-phage	one that eats	macrophage	[general]
-cytosis	increase in cells	leukocytosis	[general]
-apheresis	removal; to take away	plasmapheresis	[general]
-phylaxis	protection	prophylaxis	[general]

*Note: Handout lists "Demat" for skin — this is a typo; correct form is **derm/o** (from derm/at). Not blood/immune but flagged per course note.*

2. Blood Terms

Formed Elements & Key Components

Term	Meaning
erythrocyte (RBC)	Red blood cell; carries O_2 via hemoglobin/iron. Bone-marrow product.
leukocyte (WBC)	White blood cell; defends against infection/foreign material. Bone-marrow product.
thrombocyte (PLT)	Platelet; primary hemostasis (platelet plug). Bone-marrow product.
hemoglobin	Iron-containing protein inside RBCs that binds and carries O_2.
hematocrit	Percentage of total blood volume occupied by RBCs.
fibrinogen	Plasma protein; converted to fibrin during coagulation; also an inflammatory marker (helps locate pathogens).
bilirubin	Breakdown product of hemoglobin; transported to liver by albumin; excreted in bile; gives stool its brownish-yellow color.

Disorders & Conditions

Term	Meaning
anemia	Lower than normal RBCs or hemoglobin → tissues get less O_2. Symptoms: fatigue, weakness, pallor, dyspnea, tachycardia, dizziness, chest pain.
microcytic hypochromic anemia	Small (microcytic), pale (hypochromic) RBCs — iron deficiency. Low MCV + low MCHC on CBC.
macrocytic / megaloblastic anemia	Large, immature RBCs — B_{12} (cobalamin) or B_9 (folic acid) deficiency. Needed for DNA synthesis in rapidly dividing marrow.
pernicious anemia	B_{12} deficiency caused specifically by lack of intrinsic factor (IF) → B_{12} cannot be absorbed. *Only* use this name when IF is the problem.
aplastic anemia	Bone marrow failure; marrow not producing cell lines. Causes: radiation, toxic drugs/chemicals, marrow cancer.
hemolytic anemia	Excessive or premature RBC destruction (hemolysis). Causes: sickle cell, toxins, burns, mechanical valves, autoimmune attack. Breakdown of Hb → bilirubin → jaundice.
blood-loss anemia	Acute: normocytic/normochromic (whole blood lost). Chronic GI bleed: can become microcytic/hypochromic once iron stores are spent.
polycythemia	Too many RBCs. Causes: marrow disorder (polycythemia vera = blood cancer) or chronic hypoxia (smoking, COPD, high altitude). Effects: ↑ viscosity, ↑ volume.
leukocytosis	Increase in WBCs. Can be normal (post-op, MI) — not always infection. Monitor trends.
leukopenia	Decrease in WBCs.

neutropenia	Decrease in neutrophils (a type of leukopenia).
thrombocytopenia	Decrease in platelets.
thrombocytosis	Increase in platelets.
hemolysis	Destruction/bursting of RBCs (e.g., wrong-type transfusion: antigen meets its antibody → hemolysis).
hemostasis	Process of stopping bleeding. 5 stages: (1) vessel spasm/vasoconstriction, (2) platelet plug (primary hemostasis), (3) coagulation cascade (fibrinogen→fibrin; secondary hemostasis), (4) clot retraction, (5) fibrinolysis.
coagulation	Clot formation; cascade of clotting factors converting fibrinogen to fibrin.
embolus	Circulating clot/foreign material that lodges in and obstructs a vessel.
thrombus	Clot that forms within a vessel in situ (in place).
hemophilia	Inherited bleeding disorder; deficiency of a clotting factor → impaired coagulation.
hemochromatosis	Iron overload disorder (hemo- + chroma- + -osis); excess iron deposits in organs.
leukemia	Cancer of white blood cells; uncontrolled proliferation of abnormal WBCs in marrow/blood.
hypovolemia	Low blood volume (hypo = low, vol = volume, -emia = blood). Leads to ↓ BP → shock if severe. Observe: tachycardia *before* hypotension.
phlebotomy	Incision or puncture into a vein (blood draw).

Lab & Diagnostic Terms

Term	Meaning
CBC (complete blood count)	Reports hemoglobin, RBC count, platelets, WBCs, plus indices (MCV, MCHC).
MCV (mean corpuscular volume)	Size of the RBC. Low in microcytic anemia.
MCHC (mean corpuscular hemoglobin concentration)	How much Hb is packed in each cell. Low in hypochromic anemia.
differential (WBC differential)	Breakdown of WBC types: neutrophils, lymphocytes, monocytes, eosinophils, basophils.
ESR (erythrocyte sedimentation rate)	Inflammatory markers stick to RBCs → heavier → sediment faster. ↑ ESR = inflammation. *One of 3 markers your Patho instructor expects you to know.*
CRP (C-reactive protein)	Liver-made protein; inflammatory marker; regulates cytokines. *One of 3 markers expected.*
left shift	**Increase in immature neutrophils (bands).** Indicates marrow is releasing cells early. **NOT the same as neutrophilia** (which is simply an increase in total neutrophils).
reticulocyte count	Index of marrow activity. Reticulocytes = slightly immature RBCs (~1% of circulating). Low = lazy/blocked marrow or missing ingredients. High = marrow working hard (e.g., after hemorrhage).

3. Immune & Lymphatic Terms

Lymphatic Structures & Disorders

Term	Meaning
lymphadenopathy	Disease of lymph nodes (enlarged, abnormal nodes) — lymphaden/o + -pathy.
lymphadenitis	Inflammation of lymph nodes — lymphaden/o + -itis.
splenomegaly	Enlargement of the spleen — splen/o + -megaly.
splenectomy	Surgical removal of the spleen — splen/o + -ectomy.
lymphedema	Swelling (edema) caused by impaired lymph drainage.
lymphoma	Tumor (cancer) of lymphatic tissue — lymph/o + -oma.

Primary lymph organs: bone marrow & thymus (where immune cells are produced/matured). Secondary: spleen, lymph nodes, tonsils, appendix, Peyer's patches (GI), MALT (mucosa-associated lymphoid tissue). Losing a secondary organ (e.g., tonsils) is not the same as losing bone marrow.

Antigen vs Antibody

Antigen = a marker (surface molecule) on a cell membrane used to determine self vs non-self. **Not automatically bad** — self-antigens are left alone; foreign antigens are targeted. Also called antigenic determinant sites / epitopes.

Antibody = a protein (immunoglobulin) that the immune system *makes* to bind and neutralize a foreign antigen. Antigen + antibody = antigen-antibody (immune) complex.

Immunoglobulins (Antibody Classes)

Class	Key Facts
IgG	Most abundant. *Only Ig that crosses the placenta* — gives newborns protection for months. Long-lasting.
IgA	Secretory; in tears, saliva, GI & respiratory secretions. Protects mucous membranes (portals of entry).
IgM	First antibody to appear; suggests *current/recent* infection. Does not persist long.
IgE	Allergic reactions; defends against parasites. Mediates Type I hypersensitivity.
IgD	On B-cell membrane; stimulates B-cell proliferation when its antigen binds.

Immune Dysfunction

Term	Meaning
immunodeficiency	One or more branches of the immune system not working. Results: persistent/hard-to-treat infections, opportunistic infections, ↑ cancer incidence. Types: B-cell, T-cell, combined T+B, complement deficiency, phagocytosis disorders.
autoimmune / autoimmunity	Breakdown in self-tolerance; immune system attacks self-antigens → damage to self tissue. Often chronic with remissions and flares. Autoantibodies (e.g., ANA) are produced.

Hypersensitivity Types I–IV — Mnemonic ABCD

A = **A**llergies · **B** = anti**B**odies (cytotoxic) · **C** = immune **C**omplex · **D** = **D**elayed

Type	Mechanism	Timing	Examples
I — Allergies	IgE-mediated; mast cells/basophils degranulate → histamine, leukotrienes, bradykinin	~15 min	Allergic rhinitis (hay fever), asthma, urticaria (hives), atopic dermatitis (eczema), anaphylaxis
II — antiBodies (cytotoxic)	Antibody-mediated (IgG/IgM); antibody binds antigen on cell → binding itself damages/destroys cell	15–30 min	Blood transfusion reactions (e.g., type A recipient given type B blood)
III — immune Complex	Immune complex-mediated (IgG/IgM); complexes deposit in tissue → activate complement → inflammation	~6 hours	SLE (ANA binds DNA → complexes deposit in collagen-rich tissue)
IV — Delayed	T cell-mediated (no antibody); T cells sensitized → attack on re-exposure	1–3 days	Contact dermatitis (poison ivy), type 1 diabetes, certain transplant rejections

Key Immune / Allergy / Inflammation Terms

Term	Meaning
anaphylaxis	Most severe Type I reaction. Systemic degranulation → whole-body vasodilation → hypotension → can progress to anaphylactic shock. Also bronchoconstriction. Emergency: IM epinephrine first.
urticaria	Hives; raised, itchy skin wheals (like mosquito bites). Type I, IgE-mediated.
angioedema	Swelling of tongue, lips, part of face. Can accompany anaphylaxis or occur alone.
SLE (systemic lupus erythematosus)	Autoimmune; antigen-antibody complexes deposit in collagen-rich tissues → multi-system. Classic: butterfly/malar rash (worsened by sun). Dx = diagnosis of exclusion ("the great imitator").
ANA (antinuclear antibody)	Autoantibody directed against nuclear/DNA antigens. Primary autoantibody in SLE.
MHC (major histocompatibility complex)	Cell surface molecules (Class I & Class II) used for self-recognition. Part of self-tolerance maintenance.
degranulation	Release of inflammatory mediators (histamine, leukotrienes, bradykinin) from mast cell/basophil granules. = inflammation.
mast cells	Tissue cells similar to basophils; granules hold mediators of inflammation.
exudate	Fluid leaking from capillaries during inflammation. Types: serous (clear), serosanguineous (pink), purulent (pus). Usually beneficial — brings nutrients, O_2, antibodies.
acute-phase response	Systemic effects of inflammation: fever, leukocytosis, ↑ inflammatory markers (CRP, fibrinogen, ESR), skeletal muscle catabolism.
self-tolerance	Absence of immune response against one's own antigens. Maintained by: (1) regulatory T cells, (2) bone marrow/thymus screening of immature WBCs, (3) MHC molecules.

HIV/AIDS & Related

Term	Meaning
HIV (human immunodeficiency virus)	Retrovirus that destroys $CD4^+$ helper T cells → progressive loss of adaptive immunity.
AIDS (acquired immunodeficiency syndrome)	Advanced stage of HIV infection; defined by CD4 count < 200 cells/µL or presence of opportunistic infections / defining cancers.
CD4	Cluster of differentiation 4; membrane marker on helper T cells ("master switch"). Primary target of HIV.
opportunistic infection	Infection caused by organisms that typically do *not* cause disease in immunocompetent people; occurs when immune defenses are weakened (e.g., AIDS).
Kaposi sarcoma	Vascular tumor (associated with HHV-8); defining AIDS cancer. Presents as purple/red skin lesions.

Phagocytes & Innate Immune Cells

Term	Meaning
phagocyte	Cell that engulfs and digests foreign particles, tumor cells, virus-infected cells. Includes neutrophils and monocyte-macrophages.
macrophage	Large phagocytic cell (monocyte-macrophage lineage). Engulfs debris and acts as antigen-presenting cell (APC).
natural killer (NK) cells	Innate lymphocyte; kills tumor and virus-infected cells by releasing toxins (not phagocytosis). Does *not* need helper T activation.
cytokines	Hormone-like proteins released by WBCs to regulate immune response. Examples: interferons, interleukins, TNF, colony-stimulating factor. Can produce fever (act on hypothalamic thermoregulatory center).
complement system	Cascade of ~20 plasma proteins. Enhances phagocytosis/antibodies. Key results: degranulation, chemotaxis, lysis, opsonization. Key word to remember: **inflammation**.

4. Patho Vocabulary Tie-Ins

Terms from Pathophysiology lectures — Blood (Hematopoietic), Immune 1, Immune 2. One-line meaning; class noted. Follow these definitions for the quiz.

Term	Meaning	Class
hematopoiesis	Formation of blood cells from hematopoietic stem cells in red bone marrow.	Blood
erythropoiesis	Formation of RBCs. Stimulus: hypoxia → kidney makes EPO → marrow makes RBCs.	Blood
erythropoietin (EPO)	Kidney hormone; stimulates RBC production in response to hypoxia.	Blood
reticulocyte	Slightly immature RBC released from marrow (~1% of circulating RBCs). Retic count = index of marrow activity.	Blood

normocytic normochromic	Normal size, normal color RBCs — seen in acute blood loss (whole blood lost).	Blood
microcytic hypochromic	Small, pale RBCs — iron deficiency anemia. Low MCV + low MCHC.	Blood
macrocytic / megaloblastic	Large, immature RBCs — B_{12} or folate deficiency.	Blood
intrinsic factor (IF)	Made by gastric parietal cells; required for B_{12} absorption. Lack of IF → pernicious anemia.	Blood
pernicious anemia	B_{12} deficiency *specifically* due to lack of IF. Not all B_{12} deficiency = pernicious.	Blood
polycythemia vera	Blood cancer; marrow overproduces RBCs. One cause of polycythemia.	Blood
hypovolemia	Low blood volume. Observe: tachycardia first, then hypotension.	Blood
MCV	Mean corpuscular volume — size of RBC.	Blood
MCHC	Mean corpuscular hemoglobin concentration — Hb per cell.	Blood
bilirubin	Hb breakdown product; transported to liver by albumin; excreted in bile.	Blood
cheilosis / cheilitis	Cracked, inflamed corners of the mouth — clue for B-vitamin or iron deficiency.	Blood
paresthesias	Numbness, tingling — B_{12} deficiency neuro change.	Blood
ataxia	Coordination problems — B_{12} deficiency neuro change.	Blood
left shift	**Increase in immature neutrophils/bands.** NOT the same as neutrophilia.	Blood
innate immunity	Natural/nonspecific; present at birth. Barriers, phagocytes, NK cells, cytokines, complement.	Immune 1
adaptive immunity	Specific/acquired; learned. T cells (cell-mediated) & B cells (humoral/antibody).	Immune 1
antigen	Marker on cell membrane for self vs non-self. Not inherently bad.	Immune 1
antibody	Protein made by immune system against a foreign antigen.	Immune 1
phagocytosis	Engulfing and digesting foreign particles/dead cells.	Immune 1
APC (antigen-presenting cell)	Processes foreign antigen and displays it for T cells.	Immune 1
helper T (CD4)	"Master switch" — activates other T cells and B cells via cytokines.	Immune 1
cytotoxic T (CD8)	Kills infected/tumor cells by releasing toxins. Must be activated by helper T first.	Immune 1
regulatory/suppressor T	Shuts off immune response; suppresses self-reactive WBCs.	Immune 1
opsonization	Coating antigen-antibody complexes with complement → easier phagocytosis.	Immune 1

active immunity	Your immune system generates the response (vaccine = antigen → you make antibody). Slow onset, long-lasting.	Immune 1
passive immunity	Received antibodies/immune cells from another source (gamma globulin = antibody). Immediate, short-lived.	Immune 1
self-tolerance	No immune response against own antigens. Maintained by: regulatory T cells, bone marrow/thymus screening, MHC molecules.	Immune 2
autoantibody	Antibody directed against self-antigens (e.g., ANA in SLE).	Immune 2
hypersensitivity	Excessive immune response to an antigen most people would not react to → unnecessary inflammation/damage.	Immune 2
sensitization	Process of becoming reactive to an allergen after first exposure.	Immune 2
plasmapheresis	Removal of plasma; used as intervention for autoimmune disease (removes autoantibodies).	Immune 2
CRP	C-reactive protein; liver-made inflammatory marker.	Immune 2
ESR	Erythrocyte sedimentation rate; markers stick to RBCs → sediment faster. Inflammation marker.	Immune 2
skeletal muscle catabolism	Breakdown of muscle protein → amino acids for antibody production/tissue repair.	Immune 2
leukocytosis	Increase in WBCs — part of acute-phase response. Not always infection (post-op, MI).	Immune 2

5. Word-Building Drills (25 Terms)

Cover the right two columns and break each one down yourself.

#	Term	Breakdown	Meaning
1	**hemolysis**	hem/o (blood) + -lysis (destruction)	destruction of red blood cells
2	**erythrocyte**	erythr/o (red) + -cyte (cell)	red blood cell
3	**leukocytosis**	leuk/o (white) + cyt (cell) + -osis (abnormal condition)	abnormal increase in WBCs
4	**leukopenia**	leuk/o (white) + -penia (deficiency)	decrease in WBCs
5	**thrombocytopenia**	thromb/o (clot) + cyt/o (cell) + -penia	decrease in platelets
6	**pancytopenia**	pan- (all) + cyt/o (cell) + -penia	decrease in all blood cell types
7	**anemia**	an- (without) + -emia (blood condition)	lack of RBCs/hemoglobin
8	**polycythemia**	poly- (many) + cyt (cell) + hem (blood) + -ia	too many RBCs
9	**hemostasis**	hem/o (blood) + -stasis (stopping)	stopping of bleeding
10	**hematopoiesis**	hemat/o (blood) + -poiesis (formation)	blood cell formation
11	**erythropoiesis**	erythr/o (red) + -poiesis	RBC formation

12	**phlebotomy**	phleb/o (vein) + -tomy (incision)	incision/puncture of a vein to draw blood
13	**hemorrhage**	hem/o (blood) + -rrhage (bursting forth)	excessive bleeding
14	**neutropenia**	neutr/o (neutral) + -penia	decrease in neutrophils
15	**eosinophilia**	eosin/o (rose dye) + -philia (attraction)	increase in eosinophils
16	**anisocytosis**	aniso- (unequal) + cyt (cell) + -osis	RBCs of unequal size
17	**poikilocytosis**	poikil/o (irregular) + cyt (cell) + -osis	RBCs of irregular shape
18	**lymphadenopathy**	lymph/o + aden/o (gland) + -pathy (disease)	disease/enlargement of lymph nodes
19	**lymphadenitis**	lymphaden/o (lymph node) + -itis	inflammation of lymph nodes
20	**splenomegaly**	splen/o (spleen) + -megaly (enlargement)	enlarged spleen
21	**splenectomy**	splen/o + -ectomy (surgical removal)	removal of the spleen
22	**thymectomy**	thym/o (thymus) + -ectomy	removal of the thymus
23	**lymphoma**	lymph/o + -oma (tumor)	tumor of lymphatic tissue
24	**immunodeficiency**	immun/o (immune) + deficiency	immune system not working fully
25	**phagocyte**	phag/o (eat) + -cyte (cell)	cell that engulfs and digests

6. Common Test Traps

- Spell **erythr/o** with both r's and the h: erythrocyte. The handout's "Eryhr" is a typo. Spelling counts on this quiz.
- **-penia** = too few; **-cytosis** / **-philia** = too many. Leukopenia ≠ leukocytosis.
- **-emia** = blood condition (anemia, polycythemia, leukemia). Leukemia is a cancer, not just high WBCs.
- **-lysis** = breakdown (hemolysis); **-stasis** = stopping (hemostasis). Easy to swap.
- **-poiesis** spelling: p-o-i-e-s-i-s (hematopoiesis, erythropoiesis).
- **-tomy** = cutting into (phlebotomy); **-ectomy** = removal (splenectomy); **-ostomy** = making an opening.
- **lymphaden/o** = lymph *node*; **lymph/o** = lymph. Lymphadenopathy is about nodes.
- **Antigen** = the marker on the cell; **antibody** = the protein made against it. Her #1 mix-up.
- **thromb/o** means clot; thrombocyte = platelet. **Thrombus** stays put; **embolus** travels.
- **myel/o** = bone marrow in blood terms (spinal cord in nervous-system terms).
- **Left shift** = more immature neutrophils (bands). It is *not* neutrophilia.
- Hypersensitivity order is **ABCD**: I Allergies (IgE), II antiBodies, III immune Complex, IV Delayed (T cell).
- **Pernicious** anemia only when intrinsic factor is missing. Not every B_{12} deficiency counts.
- **Auto-** = self (autoimmune); **anti-** = against (antibody); **ante-** = before. Don't mix them up.
- **hem/o** and **hemat/o** both mean blood: hematocrit, hematoma, hemoglobin.